Currency Converter

Combined test cases | R1-R22

22 combined test designs using the required individual test-case template.

Each test-data row is a separate execution. The C in each ID identifies a combined case. All cases are drafts and have not been executed.

Controlled-result cases require a fixture that exercises the actual assessment path. TC-R22-C02 remains dependent on PD review of field-correctness expectations.

This PDF reproduces the combined test cases and requirement mapping from the project discussion.

OpenAI Codex assisted with test design and PDF preparation.

TC-R1-C01 - Reject zero and negative amounts

TC ID: TC-R1-C01

Title: Reject zero and negative amounts.

Requirement(s): R1

Type: Negative / Boundary

Priority: High

Preconditions: ConversionAssessor.validate accepts numeric double inputs directly.

Test data: Source = "USD"; target = "EUR".

Amount	Purpose
0.0	Exact rejection boundary
-0.0	Negative-zero edge case
-100.0	Negative whole number
-0.01	Negative fraction
-Double.MIN_VALUE	Negative value closest to zero
-Double.MAX_VALUE	Most negative finite value

Steps:

1. Supply the amount and valid currency codes.
2. Run input validation.
3. Capture the exception.
4. Repeat independently for each row.

Expected result: Every row throws IllegalArgumentException.

TC-R1-C02 - Accept positive amounts during validation

TC ID: TC-R1-C02

Title: Accept positive amounts during validation.

Requirement(s): R1

Type: Positive / Boundary

Priority: High

Preconditions: Numeric validation can be tested separately from GUI formatting and conversion arithmetic.

Test data: Source = "USD"; target = "EUR".

Amount	Purpose
Double.MIN_VALUE	Smallest positive value
100.0	Typical positive value
Double.MAX_VALUE	Largest finite positive value

Steps:

1. Supply the amount directly, without formatting it.
2. Run input validation.
3. Check whether an exception occurs.
4. Repeat independently for each row.

Expected result: All amounts pass validation. This does not establish that later conversion or display operations succeed.

TC-R2-C01 - Reject missing or empty currency codes

TC ID: TC-R2-C01

Title: Reject missing or empty currency codes.

Requirement(s): R2

Type: Negative

Priority: High

Preconditions: Validation accepts Java null and empty-string arguments directly.

Test data: Amount = 100.0.

Source	Target
null	"EUR"
""	"EUR"
"USD"	null
"USD"	""
null	null
""	""

Steps:

1. Assign the source and target exactly as listed.
2. Run input validation.
3. Capture the exception.
4. Repeat independently for each row.

Expected result: Every row throws IllegalArgumentException. Here, null is not the text "null".

TC-R2-C02 - Accept populated, valid currency codes

TC ID: TC-R2-C02

Title: Accept populated, valid currency codes.

Requirement(s): R2

Type: Positive

Priority: High

Preconditions: Input validation is available.

Test data: Amount = 100.0.

Source	Target
"USD"	"EUR"
"USD"	"USD"

Steps:

1. Supply the amount and currency codes.
2. Run input validation.
3. Check whether an exception occurs.

Expected result: Both rows pass validation, including the same-currency pair.

TC-R3-C01 - Enter amounts and verify their returned values

TC ID: TC-R3-C01

Title: Enter amounts and verify their returned values.

Requirement(s): R3, R7, R13, R16

Type: Positive / Edge

Priority: High

Preconditions: Converter is open with a dot-decimal locale. Source and target are both USD.

Test data:

Entered text	Expected inputAmount	Expected displayed amount / conversionResult
"100"	100.0	"100.00" / 100.0
"12.50"	12.5	"12.50" / 12.5
"0.01"	0.01	"0.01" / 0.01
"1000000"	1000000.0	"1000000.00" / 1000000.0
"1.234567"	1.234567	"1.23" / 1.23
"12.3"	12.3	"12.30" / 12.3
"1.234"	1.234	"1.23" / 1.23
"12.34"	12.34	"12.34" / 12.34

Steps:

1. Clear the amount field and enter the listed text without quotation marks.
2. Confirm the field preserves the entered text before invoking assessment.
3. Run assessment using the corresponding numeric amount.
4. Capture the baseline displayed result and exchange rate.
5. Compare extracted values with their UI labels.
6. Inspect inputAmount and conversionResult.
7. Close or reset the converter.

Expected result: Entry text is correct; inputAmount retains the original numeric value; conversionResult matches the displayed rounded amount. Extracted rate is 1.0.

TC-R3-C02 - Replace or edit an amount without returning stale values

TC ID: TC-R3-C02

Title: Replace or edit an amount without returning stale values.

Requirement(s): R3, R7, R13, R16

Type: Positive

Priority: High

Preconditions: Converter is open with USD to USD and a dot-decimal locale.

Test data:

Starting text	Action	Final text
"100"	Replace the entire value	"250"
"125"	Delete the last digit and type 0	"120"
"100"	Replace the entire value	"250.75"

Steps:

1. Enter the starting text and establish its conversion.
2. Perform the listed edit.
3. Verify the final field text.
4. Assess the final numeric amount and inspect the returned record.
5. Compare extracted and returned results with the final display.
6. Close or reset the converter.

Expected result: Final text matches the table. Returned input and conversion values correspond to the final amount, with no stale starting value.

TC-R4-C01 - Select each source currency and verify its returned code

TC ID: TC-R4-C01

Title: Select each source currency and verify its returned code.

Requirement(s): R4, R14

Type: Positive

Priority: High

Preconditions: Converter is open; amount = "100"; target = EUR. Start with a different source from the one being tested.

Test data: Source selections: USD, EUR, GBP, JPY, CAD, AUD, CHF. Execute once for each currency.

Steps:

1. Focus the source dropdown.
2. Press Enter, select the currency using arrow keys, and confirm with Enter.
3. Read the selected source directly from the dropdown.
4. Run assessment with that source and target EUR.
5. Inspect sourceCurrency.
6. Reset before testing the next currency.

Expected result: Each currency can be selected. The dropdown and returned sourceCurrency both match the selected code. Amount remains 100 and target remains EUR.

TC-R4-C02 - Preserve the latest source selection through changes and focus movement

TC ID: TC-R4-C02

Title: Preserve the latest source selection through changes and focus movement.

Requirement(s): R4, R14

Type: Positive

Priority: High

Preconditions: Converter is open; amount = "100"; target = EUR; initial source = CHF.

Test data:

Selection sequence	Expected final source
EUR -> GBP -> JPY	JPY
GBP -> EUR	EUR

Steps:

1. Select each source in the listed order.
2. After each selection, press Tab to leave the dropdown.
3. Verify the selection persists, then return focus for the next choice.
4. Assess the final selection.
5. Inspect sourceCurrency.

Expected result: Every selection persists after focus moves. The final dropdown selection and returned field match the table, without retaining an earlier code.

TC-R5-C01 - Select each target currency and verify its returned code

TC ID: TC-R5-C01

Title: Select each target currency and verify its returned code.

Requirement(s): R5, R15

Type: Positive

Priority: High

Preconditions: Converter is open; amount = "100"; source = USD. Start with a different target from the one being tested.

Test data: Target selections: USD, EUR, GBP, JPY, CAD, AUD, CHF. Execute once for each currency.

Steps:

1. Focus the target dropdown.
2. Press Enter, select the currency using arrow keys, and confirm with Enter.
3. Read the selected target directly from the dropdown.
4. Run assessment with source USD and that target.
5. Inspect targetCurrency.
6. Reset before testing the next currency.

Expected result: Each currency can be selected. The dropdown and returned targetCurrency match the selected code. Amount remains 100 and source remains USD.

TC-R5-C02 - Preserve the latest target selection through changes and focus movement

TC ID: TC-R5-C02

Title: Preserve the latest target selection through changes and focus movement.

Requirement(s): R5, R15

Type: Positive

Priority: High

Preconditions: Converter is open; amount = "100"; source = USD; initial target = CHF.

Test data:

Selection sequence	Expected final target
EUR -> GBP -> JPY	JPY
GBP -> EUR	EUR

Steps:

1. Select each target in the listed order.
2. After each selection, press Tab to leave the dropdown.
3. Verify the selection persists, then return focus for the next choice.
4. Assess the final selection.
5. Inspect targetCurrency.

Expected result: Every selection persists after focus moves. The final dropdown selection and returned field match the table.

TC-R6-C01 - Wait successfully for valid and recovered results

TC ID: TC-R6-C01

Title: Wait successfully for valid and recovered results.

Requirement(s): R6

Type: Positive / Edge

Priority: High

Preconditions: Converter and waitForResult are available. Default timeout is five seconds.

Test data:

Initial amount	Final amount	Source to target
Empty	100	USD to EUR
Empty	100	USD to USD
0	100	USD to EUR

Steps:

1. Set the currency pair and initial amount.
2. Confirm the result container is hidden.
3. Enter the final amount.
4. Call waitForResult.
5. Once visible, call waitForResult again without changing anything.

Expected result: Both calls return true. The second succeeds when the container is already visible, without requiring another visibility transition.

TC-R6-C02 - Handle delayed and missing results

TC ID: TC-R6-C02

Title: Handle delayed and missing results.

Requirement(s): R6

Type: Boundary / Negative

Priority: High

Preconditions: A controlled fixture can schedule visibility independently of the wait. Timeout is five seconds.

Test data: Amount = 100; USD to EUR.

Visibility behavior	Expected wait outcome
Hidden initially; visible after one second	true after visibility
Hidden throughout the timeout	false after timeout

Steps:

1. Configure the listed visibility behavior.
2. Start waiting while the container is hidden.
3. Record the return value, elapsed time, and final visibility.
4. Reset the fixture between rows.

Expected result: The wait succeeds only after visibility occurs. A missing result returns false around the configured deadline, allowing normal scheduling overhead, and does not hang.

TC-R7-C01 - Extract displayed amounts and exchange rates correctly

TC ID: TC-R7-C01

Title: Extract displayed amounts and exchange rates correctly.

Requirement(s): R7, R16

Type: Positive

Priority: High

Preconditions: Converter is open with a dot-decimal locale and the supplied fixed currency rates.

Test data: Amount = 100.

Source to target	Expected amount text	Expected rate text
USD to EUR	"86.26"	"0.862623"
USD to USD	"100.00"	"1.000000"
JPY to GBP	"0.47"	"0.004707"

Steps:

1. Assess the listed conversion.
2. Capture the baseline amount and rate labels before the temporary update probe.
3. Extract and parse both displayed values.
4. Compare extracted values with the recorded labels.
5. Inspect conversionResult.

Expected result: Displayed strings match the table. Extraction retains all displayed numeric digits, and conversionResult matches the baseline displayed amount.

TC-R7-C02 - Extract current values after an input change

TC ID: TC-R7-C02

Title: Extract current values after an input change.

Requirement(s): R7, R16

Type: Positive

Priority: High

Preconditions: Establish a visible conversion for 100, USD to USD.

Test data:

Change	Expected amount	Expected rate
Change amount to 200	200.0	1.0
Keep amount 100; change target to EUR	86.26	0.862623

Steps:

1. Restore the starting conversion before each row.
2. Make the listed change.
3. Wait until the displayed contents reflect the change.
4. Extract the amount and rate.
5. Assess the final inputs and inspect conversionResult.

Expected result: Extracted values and the returned conversion amount match the updated conversion. No earlier amount or rate is returned.

TC-R8-C01 - Format input and return the correct input decimal count

TC ID: TC-R8-C01

Title: Format input and return the correct input decimal count.

Requirement(s): R8, R9, R13, R17

Type: Positive / Boundary / Edge

Priority: High

Preconditions: Dot-decimal locale is fixed. The assessment's formatted-input string can be observed through a test fixture or debugger. Use USD to USD.

Test data:

Numeric input	Expected formatted string
100.0	"100.00"
12.5	"12.50"
12.34	"12.34"
12.344	"12.34"
12.346	"12.35"
1.125	"1.13"
9.999	"10.00"
0.001	"0.00"
1.234	"1.23"
1000000.0	"1000000.00"

Steps:

1. Assess the original numeric input.
2. Capture the formatted input produced by String.format("%.2f", amount).
3. Compare it with the expected string.
4. Count its fractional digits independently.
5. Inspect inputDecimalPlaces and inputAmount.

Expected result: The formatted string matches the table; its count and inputDecimalPlaces are both 2. inputAmount retains the original numeric input, including digits removed by formatting.

TC-R9-C01 - Count result decimals and return truthful precision fields

TC ID: TC-R9-C01

Title: Count result decimals and return truthful precision fields.

Requirement(s): R9, R11, R18, R20, R22

Type: Positive / Negative / Boundary

Priority: High

Preconditions: A controlled assessment fixture supplies exact result strings. Visibility, accuracy, and dynamic update/restoration pass independently. Use valid USD to USD codes and a fixed dot-decimal locale.

Test data:

Input	Result text	Result decimal count	Precision flag
100	"92.35"	2	true
12.5	"11.50"	2	true
1.234	"1.1400"	4	true
100	"92.350"	3	true
100	"92.3"	1	false
100	"92"	0	false
0.001	"0.0001"	4	true
1	"0.00"	2	true

Steps:

1. Configure the baseline result string exactly as listed.
2. Supply a controlled rate that makes baseline accuracy pass.
3. Arrange a changed result distinct from baseline and an exact restoration.
4. Run the actual assessment path.
5. Count decimals independently in the formatted input and baseline result.
6. Inspect resultDecimalPlaces, decimalPrecisionMaintained, and conversionComplete.

Expected result: Input decimal count is 2. Result count and precision flag match the table. Because other completion conditions pass, conversionComplete equals the precision flag.

TC-R10-C01 - Validate accuracy and return truthful accuracy and completion flags

TC ID: TC-R10-C01

Title: Validate accuracy and return truthful accuracy and completion flags.

Requirement(s): R10, R19, R22

Type: Positive / Negative / Boundary

Priority: High

Preconditions: A controlled assessment fixture supplies the listed amount, displayed rate, and result. Visibility, precision, and update/restoration pass. Use valid USD to USD codes.

Test data:

Amount	Rate	Result text	Expected accurate
1	1	"1.000"	true
1	1	"1.009"	true
1	1	"0.991"	true
1	0.01	"0.00"	true
1	1	"1.011"	false
1	1	"0.989"	false

Steps:

1. Configure the fixture with the listed baseline values.
2. Arrange a distinct changed result and an exact baseline restoration.
3. Run the actual assessment path.
4. Independently calculate Math.abs(result - amount * rate).
5. Compare the difference with 0.01.
6. Inspect conversionAccurate and conversionComplete.

Expected result: Both flags match the table because all other completion conditions pass. The fourth row tests equality at the tolerance boundary without subtracting nearly equal numbers.

TC-R12-C01 - Update and restore a conversion successfully

TC ID: TC-R12-C01

Title: Update and restore a conversion successfully.

Requirement(s): R12, R21, R22

Type: Positive

Priority: High

Preconditions: Converter is open; baseline visibility, accuracy, and precision pass.

Test data:

Source to target	Original -> changed -> restored amount
USD to USD	100 -> 200 -> 100
USD to USD	12.5 -> 25 -> 12.5
USD to EUR	100 -> 200 -> 100

Steps:

1. Run assessment using the original amount and listed currencies.
2. Observe the baseline result.
3. Observe the assessor doubling the amount and obtaining a different result.
4. Observe restoration of the original amount and result.
5. Inspect resultUpdatedOnChange and conversionComplete.
6. Verify the original numeric amount and currencies are restored.

Expected result: Changed result differs from baseline; restored result equals baseline. Both flags are true, and original inputs are restored.

TC-R12-C02 - Handle decreasing and repeated amount changes

TC ID: TC-R12-C02

Title: Handle decreasing and repeated amount changes.

Requirement(s): R12

Type: Positive

Priority: High

Preconditions: Converter is open with USD to USD. Driver operations can change amounts directly.

Test data:

Amount sequence	Expected result sequence
100 -> 50 -> 100	"100.00" -> "50.00" -> "100.00"
100 -> 200 -> 300 -> 100	"100.00" -> "200.00" -> "300.00" -> "100.00"

Steps:

1. Establish and record the original amount and result.
2. Enter each subsequent amount in order.
3. Wait for the corresponding result after each change.
4. Restore the exact original amount text.
5. Check final amount, result, and currencies.

Expected result: Every result matches the latest input. Final amount and result equal their originals. These sequences are checked separately because the supplied update helper only doubles the amount.

TC-R12-C03 - Reject unsuccessful updates or restoration

TC ID: TC-R12-C03

Title: Reject unsuccessful updates or restoration.

Requirement(s): R12, R21, R22

Type: Negative / Edge

Priority: High

Preconditions: A controlled fixture can supply frozen or incorrectly restored results. Baseline visibility, precision, and accuracy pass. Use USD to USD with displayed rate 1.

Test data:

Situation	Original amount	Baseline -> changed -> restored result
Frozen result	100	"100.00" -> "100.00" -> "100.00"
Change hidden by rounding	0.001	"0.00" -> "0.00" -> "0.00"
Failed restoration	100	"100.00" -> "200.00" -> "150.00"

Steps:

1. Configure the listed result sequence; the rounding row can also use the real UI.
2. Run assessment with the original amount.
3. Confirm the probe attempts twice the amount, followed by the original amount.
4. Compare baseline, changed, and restored strings.
5. Inspect resultUpdatedOnChange and conversionComplete.
6. Close the converter to clear any failed-restoration state.

Expected result: Both flags are false. An unchanged display or unsuccessful restoration must not be reported as a successful dynamic update.

TC-R22-C01 - Reject incomplete conversions when visibility or multiple checks fail

TC ID: TC-R22-C01

Title: Reject incomplete conversions when visibility or multiple checks fail.

Requirement(s): R22

Type: Negative

Priority: High

Preconditions: A controlled assessment fixture can return visibility outcomes and parseable result strings. Amount = 100; source = USD; target = USD; rate = 1.

Test data:

Situation	Visibility	Baseline -> changed -> restored result
Initial visibility wait fails	Initial wait false; later probe waits true	"100.00" -> "200.00" -> "100.00"
Accuracy and precision both fail	true	"100.1" -> "200.1" -> "100.1"

Steps:

1. Configure the listed visibility responses and result strings.
2. Run the actual assessment path.
3. Independently verify which conditions pass and fail.
4. Inspect conversionComplete.

Expected result: conversionComplete is false for both rows. The first isolates failed visibility; the second verifies combined accuracy and precision failures.

TC-R22-C02 - Reject completion when a returned currency disagrees with the selection

TC ID: TC-R22-C02

Title: Reject completion when a returned currency disagrees with the selection.

Requirement(s): R14, R22

Type: Negative

Priority: High

Preconditions: PD review confirms that R22 includes returned-field correctness. A controlled selection fault can make the actual source differ from the requested source before the record is created.

Test data: Amount = 100; requested source = USD; actual selected source = EUR; target = USD. Controlled visibility, accuracy, precision, and update checks all pass.

Steps:

1. Configure the source-selection fault.
2. Run assessment requesting USD to USD.
3. Read the actual source selection independently.
4. Compare it with returned sourceCurrency.
5. Inspect conversionComplete.

Expected result: Under the stated interpretation, a record with an incorrect source field must not report successful completion. Record any contrary behavior as a defect. This case remains review-dependent because the current completion calculation does not explicitly check field correctness.

Requirement mapping

Requirement	Combined cases
R1	TC-R1-C01, TC-R1-C02
R2	TC-R2-C01, TC-R2-C02
R3	TC-R3-C01, TC-R3-C02
R4	TC-R4-C01, TC-R4-C02
R5	TC-R5-C01, TC-R5-C02
R6	TC-R6-C01, TC-R6-C02
R7	TC-R3-C01, TC-R3-C02, TC-R7-C01, TC-R7-C02
R8	TC-R8-C01
R9	TC-R8-C01, TC-R9-C01
R10	TC-R10-C01
R11	TC-R9-C01
R12	TC-R12-C01, TC-R12-C02, TC-R12-C03
R13	TC-R3-C01, TC-R3-C02, TC-R8-C01
R14	TC-R4-C01, TC-R4-C02, TC-R22-C02
R15	TC-R5-C01, TC-R5-C02
R16	TC-R3-C01, TC-R3-C02, TC-R7-C01, TC-R7-C02
R17	TC-R8-C01
R18	TC-R9-C01
R19	TC-R10-C01
R20	TC-R9-C01
R21	TC-R12-C01, TC-R12-C03
R22	TC-R9-C01, TC-R10-C01, TC-R12-C01, TC-R12-C03, TC-R22-C01, TC-R22-C02